

KLYUCHI ALTAI, Russian Federation

WMO: 360210

Lat: 52.263N

Lon: 79.199E

Elev: 143

StdP: 99.62

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-31.7	-37.9	0.1	-34.9	-34.4	0.1	-31.7	8.6	-8.0	7.6	-7.6	1.5	270	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.4	19.1	30.5	18.5	28.8	17.8	21.0	28.5	20.0	26.9	19.1	26.1	3.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.3	17.6	12.9	22.5	16.5	12.0	22.0	61.4	28.3	58.1	26.9	55.1	25.9	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.4	-36.3	35.8	3.9	1.9	-39.1	37.2	-41.3	38.3	-43.5	39.3	-46.3	40.7		
(5)			-36.7	22.2	4.1	1.6	-39.6	23.3	-41.9	24.3	-44.2	25.2	-47.2	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-16.7	-14.2	-5.7	6.5	13.7	19.6	20.8	18.8	12.5	5.0	-5.2	-12.5		
	DBStd	14.67	8.61	7.80	7.14	5.80	5.17	4.02	3.25	3.83	4.73	5.22	7.95	7.77		
	HDD10.0	3498	828	679	487	133	24	1	0	1	27	168	454	697		
	HDD18.3	5604	1086	912	746	356	159	34	14	41	184	412	704	955		
	CDD10.0	1181	0	0	0	28	139	288	336	273	102	14	0	0		
	CDD18.3	242	0	0	0	1	15	71	92	55	8	0	0	0		
	CDH23.3	2664	0	0	0	21	248	767	846	647	134	2	0	0		
	CDH26.7	909	0	0	0	4	72	277	280	234	42	0	0	0		
(14) Wind	WSAvg	2.9	2.6	2.7	3.0	3.4	3.4	3.0	2.7	2.7	2.7	2.8	2.9	3.0		
(15) Precipitation	PrecAvg	279	13	12	15	16	21	35	59	29	20	21	22	17		
	PrecMax	354	24	22	46	37	75	120	114	98	55	64	48	36		
	PrecMin	148	2	0	2	0	5	2	9	0	1	5	5	8		
	PrecStd	47	6	6	10	10	15	25	24	18	12	15	11	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	2.6	12.9	26.2	32.3	34.2	34.0	35.0	31.5	22.1	11.6	2.8		
		MCWB	1.0	0.6	7.2	14.2	18.3	19.0	20.5	20.0	17.6	12.4	7.5	1.2		
	2%	DB	0.0	0.6	8.2	22.1	28.6	31.8	31.6	31.8	27.0	18.4	8.7	0.9		
		MCWB	-1.1	-0.9	4.7	12.5	16.3	17.9	20.1	18.8	15.4	10.5	5.4	-0.5		
	5%	DB	-2.3	-1.0	5.2	19.1	26.4	29.8	29.8	29.2	23.9	15.6	6.1	-0.5		
		MCWB	-3.5	-2.4	2.5	10.6	15.1	17.7	19.2	17.9	13.9	9.1	3.5	-1.7		
	10%	DB	-4.6	-3.0	3.0	16.3	23.7	27.8	28.0	27.0	21.0	12.8	3.7	-2.2		
		MCWB	-5.6	-4.1	1.2	9.2	13.9	17.2	18.5	17.2	12.6	7.7	2.1	-3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.2	7.7	15.1	19.5	20.8	23.1	22.2	18.3	13.0	8.4	1.5		
		MCDB	2.1	2.2	11.9	23.2	30.7	30.9	29.6	30.6	29.5	20.5	10.7	2.5		
	2%	WB	-1.0	-0.8	4.8	13.0	17.4	19.6	21.4	20.3	16.0	11.2	5.6	-0.1		
		MCDB	-0.1	0.4	7.9	20.9	26.8	27.9	28.1	28.1	24.9	17.1	8.0	0.7		
	5%	WB	-3.5	-2.5	2.8	11.2	15.9	18.8	20.5	19.0	14.6	9.7	3.7	-1.6		
		MCDB	-2.5	-1.2	4.9	17.9	24.3	26.6	26.8	26.9	22.3	14.8	5.9	-0.7		
	10%	WB	-5.6	-4.2	1.2	9.7	14.6	18.0	19.6	17.9	13.2	8.2	2.1	-3.2		
		MCDB	-4.6	-3.0	2.9	15.6	22.0	25.5	25.6	25.4	20.0	12.4	3.7	-2.1		
(35) Mean Daily Temperature Range	MDBR		8.5	9.3	9.1	11.7	12.5	11.8	11.0	12.2	11.7	9.4	7.3	8.2		
	5% DB	MCDBR	9.0	8.8	10.3	15.9	16.6	15.2	13.9	16.2	16.8	14.3	8.9	8.5		
		MCWBR	8.5	7.9	7.1	8.2	7.6	5.8	4.9	5.9	7.7	7.9	6.4	8.0		
	5% WB	MCDBR	9.3	8.5	9.6	14.6	14.6	12.5	11.3	13.8	14.7	12.9	8.1	8.5		
		MCWBR	8.8	7.8	6.9	8.1	7.2	5.4	4.7	5.5	6.9	7.6	6.2	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.336	0.391	0.411	0.415	0.416	0.442	0.420	0.377	0.354	0.339	0.310		
	taud		2.028	1.955	1.932	2.146	2.185	2.265	2.200	2.227	2.293	2.256	2.112	2.001		
	Ebn at Noon		648	735	771	819	837	838	807	803	796	728	598	549		
	Edh at Noon		81	117	148	136	137	128	135	124	103	87	75	70		
(44) All-Sky Solar Radiation	RadAvg		1.25	2.38	3.82	4.77	6.02	6.40	5.79	5.09	3.71	2.04	1.04	0.83		
	RadStd		0.11	0.11	0.29	0.35	0.43	0.39	0.39	0.32	0.29	0.27	0.11	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.81	-0.86	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page